ChallengeMe — Project Description

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Overview

ChallengeMe is a mobile social app where users take on AI-generated challenges alone or with friends, prove completion with a photo, and earn XP, streaks and badges. Its motto is "Don't just scroll. Challenge yourself."

The project builds a working prototype and measures whether AI personalization and adaptive difficulty increase challenge completion and retention compared with a non-AI baseline. It is designed for a team of two over one semester.

Goals and research questions

Product goals

- Let users generate, join and create challenges in 13 categories
- Support friends, groups, leaderboards and chat for each challenge
- Keep the free version fully usable, with no pay-to-win

Research questions

1. **RQ1:** Do personalized recommendations raise the challenge completion rate versus random or popularity-based ones?
2. **RQ2:** Does adaptive difficulty improve completion and 14-day retention versus fixed difficulty?
3. **RQ3:** How closely does AI photo verification agree with human raters, and are its confidence scores calibrated?

Scope

In the MVP

- Sign-up, profile, XP, levels, streaks and achievements
- Browse and join challenges; create your own with AI help
- AI challenge generator, including 7, 14 and 30-day programs
- Photo proof with AI verification, a confidence score and a dispute button
- Friends, follow, per-challenge leaderboard and chat
- Recommendations and difficulty adaptation

- Sandbox ChallengeMe+ paywall and cosmetic shop, with no real payments

Out of scope

- Real in-app purchases and app-store release
- Seasonal events, sponsored challenges and creator tips
- Moderation tools beyond a basic report button
- Video verification and push notifications

Technical architecture

The app is a React Native client talking to a stateless Python API on Google Cloud, with AI calls isolated behind service modules.

Layer	Technology	Notes
Mobile app	React Native (TypeScript) with Expo, run in Expo Go	Only Expo Go-compatible libraries: expo-camera, expo-image-picker, expo-router
API	Python 3.12, FastAPI, Pydantic	REST with an OpenAPI contract shared by both members
Hosting	Google Cloud Run	Containerized, scales to zero, fits free-tier credits
Database	Cloud SQL (PostgreSQL)	Users, challenges, tasks, completions, events
Realtime chat	Firestore	Per-challenge message collections; fallback is polling the API
Auth	Firebase Authentication	The API verifies ID tokens
File storage	Cloud Storage	Proof images via signed upload URLs
AI	Vertex AI (Gemini) or Claude API	Schema-validated JSON for generation; vision model for verification
Delivery	GitHub Actions to Cloud Build	Automatic API deploy on merge

Expo Go cannot run native purchase libraries, which is why monetization stays in sandbox mode (see Monetization).

AI components and evaluation

Each AI feature has a simple baseline, so improvement is measurable.

Component	Baseline	Proposed method	Metrics
Recommendations	Random or most popular	Content-based scoring on category, difficulty and history, plus a bandit for exploration	Completion rate, precision@5, NDCG@5
Difficulty adaptation	Fixed difficulty	Rule-based controller on the last 5 results, upgraded to a learned model if data allows	Completion rate, 14-day retention, streak length
Challenge generation	Hand-written templates	LLM with schema-validated output	Schema validity rate, user rating (1-5), edit rate before publishing
Photo verification	Human raters	Vision model returning a verdict and a confidence score	Accuracy, precision, recall, Cohen's kappa, calibration error, dispute rate

Study design. Users are randomly assigned to a control group (random recommendations, fixed difficulty) or a treatment group (full AI features) for about 3 weeks. A power analysis (d = 0.5, alpha = 0.05, power = 0.80) needs 64 users per group, so about 170 participants with 25% dropout; 40-60 users still support a descriptive evaluation. Participants are adult university students who give informed consent. Experience is measured with the System Usability Scale (SUS).

Monetization

Monetization follows a no-gambling, no-pay-to-win rule: nothing bought gives a competitive advantage, and chat and ordinary challenges stay free. It runs in sandbox mode with a fake checkout.

Offer	Price	Purpose
ChallengeMe+	$0.99/month	Unlimited AI generation, coaching and verification

Offer	Price	Purpose
30-Day Challenge Pack	$0.99 one-time	Personalized 30-day program
Cosmetics	Earned or $0.99	Profile frames, themes, team badges

The study measures paywall conversion, an A/B test of $0.99 versus $1.99, and a short survey on whether prompts felt intrusive. Sponsored challenges and creator tips are noted as future work.

Team split

The work is split 50/50 at about 130 estimated hours each, and every task has exactly one owner. Member A owns the Expo setup, the Google Cloud deployment and the AI backend; Member B owns the auth, app screens, social features and the user study. Replace A and B with your names.

Member A: Expo, cloud and AI backend

#	Task	Hours
1	Expo setup, navigation and API client	14
2	Google Cloud deployment, CI/CD, monitoring	14
3	API contract (OpenAPI) and data model	10
4	Challenge, task and progress API; XP, streak and achievement engine	18
5	AI challenge generator service	14
6	Recommendation and difficulty adaptation	20
7	Verification service with confidence score	16
8	Leaderboard API	6
9	Event logging and metrics scripts	10
10	Report chapters: architecture, backend and AI	8
	Total	**130**

Member B: auth, screens, social and study

#	Task	Hours
1	Wireframes and design system	10
2	Auth, profiles, friends and follow API	14
3	Auth, Home, Explore, challenge detail and profile screens	22
4	AI Studio screen	8
5	Camera upload and dispute UI	10
6	Leaderboard screen and challenge chat (Firestore)	18
7	Sandbox paywall and cosmetic shop	18
8	Study design, recruitment, survey and statistics	14
9	Usability testing and demo script	8
10	Report chapters: frontend, evaluation and monetization	8
	Total	**130**

Member A publishes the API contract in weeks 1-2 so Member B can build against mocks. Both members review each other's pull requests and join the demo and final presentation.

Timeline

The plan runs 14 weeks, with the pilot study in weeks 11-13 and a feature freeze at the end of week 9.

Weeks	Milestone	Member A	Member B
1-2	Foundations	API contract, DB schema, Expo app skeleton, Cloud Run skeleton	Wireframes, design system, auth API
3-5	Core app	Challenge API, XP and streak engine, Expo navigation and API client	Auth, Home, Explore, detail and profile screens
6-8	AI and social	Generator, verification, recommender v1	AI Studio, proof upload, leaderboard, chat

Weeks	Milestone	Member A	Member B
9	Adaptation and monetization	Difficulty controller, event logging	Sandbox paywall, cosmetic shop
10	Hardening	Integration tests, deployment fixes	Usability fixes, onboarding, pilot materials
11-13	Pilot study	Monitor, fix bugs, export data	Recruit users, run survey, collect feedback
14	Wrap-up	Analysis scripts, backend chapters	Statistics, frontend chapters, demo

Risks, ethics and deliverables

Risk	Mitigation
Too few pilot users	Recruit through course groups; analyze descriptively if under 64 per group
AI cost or latency	Daily per-user limits, caching, a cheaper model for generation
Photo verification errors	Show confidence, allow disputes, report error rates openly
Scope creep	Feature freeze in week 9 and the out-of-scope list above
Data privacy	Adults only, consent form, anonymized exports, GDPR-compliant retention

Deliverables

- Working app (Expo Go) and deployed API on Google Cloud
- Source repository with README and API documentation
- Evaluation report with results for RQ1-RQ3
- Final presentation and live demo